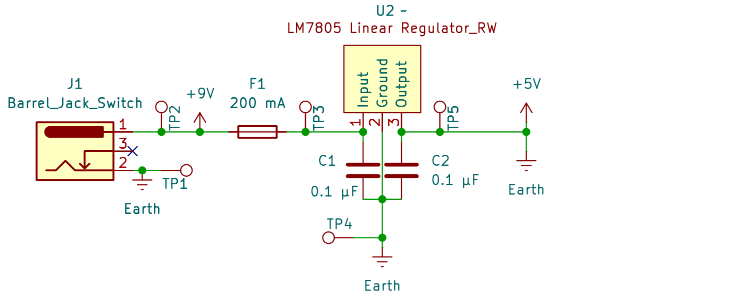
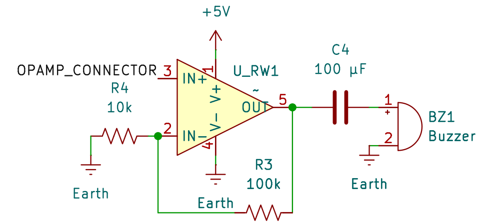
Power Supply



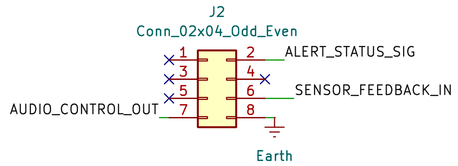
The board is powered through a 5 V DC barrel jack. The input passes through a 200–250 mA fuse for overcurrent protection and includes decoupling capacitors to filter noise and stabilize the supply. The LM7805 voltage regulator footprint is available but left unpopulated when a regulated 5 V adapter is used. A 10 µF electrolytic capacitor provides bulk decoupling, and 0.1 µF ceramic capacitors are placed near integrated circuit power pins. Optional protection components such as a Schottky diode for reverse polarity and a transient voltage suppressor can be added near the barrel jack if needed. Test points for the 5 V rail and ground are included for easy measurement and debugging.

Speaker System



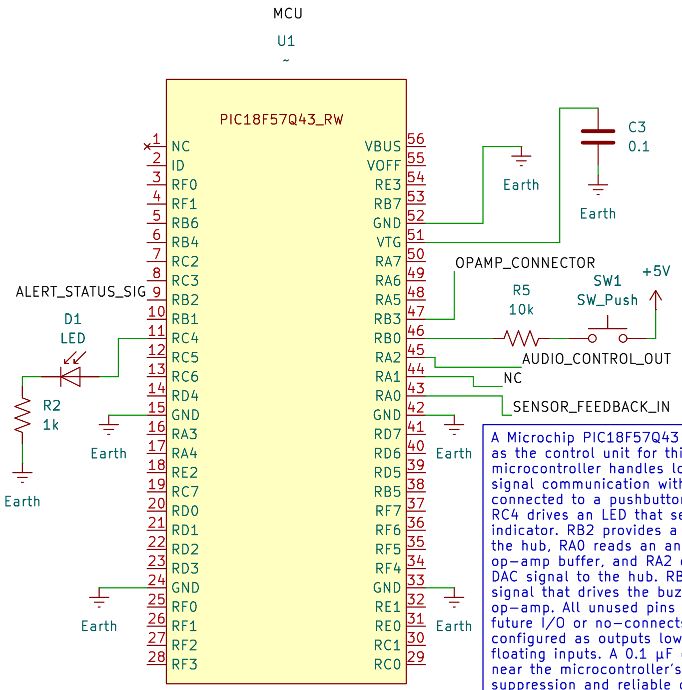
The audio circuit uses an MCP6004 operational amplifier and a WT-1205 active buzzer. The op-amp is configured as a non-inverting amplifier and buffers the PWM output from the microcontroller to properly drive the buzzer. The op-amp output feeds a transistor that switches the buzzer's negative terminal to ground, while the positive terminal connects directly to the 5 V supply. This configuration allows for clean audio output while protecting the microcontroller from high current draw. The op-amp includes a 0.1 µF decoupling capacitor near its power pins, and a 10 nF capacitor across the buzzer may be added to reduce noise or ringing.

Ribbon Connector



An 8-pin ribbon connector provides communication between this board and the hub. Pin 1 is unused, and pin 2 carries the digital alert signal to the hub. Pins 3 through 5 are reserved for future use. Pin 6 brings an analog signal into the board, routed through the op-amp buffer and read by RA0. Pin 7 carries an analog or DAC output signal from RA2 to the hub, and pin 8 is the shared system ground. Each digital line that leaves the board includes a 100 Ω series resistor to protect against transients or accidental contention. No 5 V power is shared across the ribbon.

Microcontroller



A Microchip PIC18F57Q43 Curiosity Nano is used as the control unit for this board. The microcontroller handles local input, output, and signal communication with the hub. RB0 is connected to a pushbutton used for testing, and RC4 drives an LED that serves as a local indicator. RB2 provides a digital alert signal to the hub, RA0 reads an analog input through an op-amp buffer, and RA2 outputs an analog or DAC signal to the hub. RB3 generates a PWM signal that drives the buzzer through the op-amp. All unused pins are labeled as either future I/O or no-connects and will be configured as outputs low in firmware to prevent floating inputs. A 0.1 µF capacitor is placed near the microcontroller's power pins for noise suppression and reliable operation.

General Notes

All resistors are quarter watt unless otherwise specified, and all capacitors are listed in microfarads unless otherwise noted. Power traces are designed with at least 40 mil width to safely handle the system current. Component values such as op-amp gain resistors and buzzer coupling capacitors may be adjusted during testing to achieve the desired audio performance. The board is electrically isolated from the hub except for signal and ground connections. This schematic represents version 1.0 of the Audio and Alert subsystem, completed in October 2025.

Expansion and Design Intent

Unused microcontroller ports such as RA1 and other RC, RD, or RF pins are labeled for future I/O expansion. Extra channels of the MCP6004 are available for additional analog signal conditioning. The layout and connector pinout follow the Irri-Gators team interface standard to ensure compatibility between subsystems. All unused pins are clearly marked with either "Future I/O" or "No Connect" symbols. The board's design allows for simple integration of new components or communication features without schematic restructuring.

Testing

The board includes a local pushbutton and LED for functional testing. The pushbutton, connected to RB0, uses the internal pull-up resistor of the PIC and allows manual triggering of alerts or testing of logic. The LED, connected to RC4 through a 330 Ω resistor, provides immediate visual confirmation of the microcontroller's output state. These features are intended only for debugging and do not interface with the hub.

Subsystem Description

This board serves as the Audio and Alert subsystem for the Irri-Gators project, designed by Rylee Wirt. It provides both audible and visual alerts to indicate system status or errors and connects to the main hub through an 8-pin ribbon cable. Each subsystem in the design has its own regulated 5 V power supply, and only the ground line is shared across the ribbon connection. This prevents backfeeding between supplies while maintaining a common signal reference. The schematic includes clear separation between power, control, and signal-conditioning sections to simplify troubleshooting and future expansion.

Sheet: /
File: SchematicDesign_RW.kicad_sch

Title: Speaker with Microcontroller Subsystem

Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.4		Id: 1/1